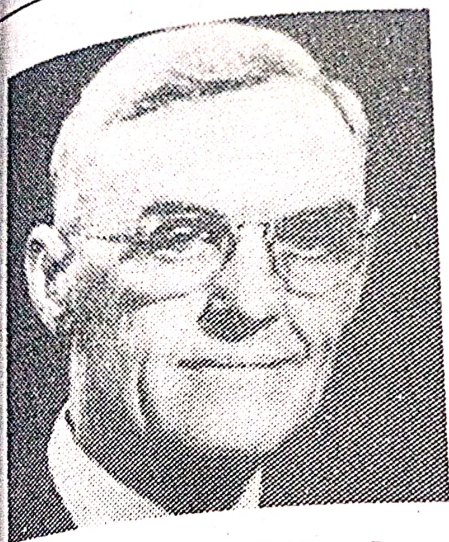


Snedecor's F-Distribution



George Waddel Snedecor (October 20, 1881 – February 15, 1974) was an American mathematician and statistician. He contributed to the foundations of analysis of variance, data analysis, experimental design, and statistical methodology. Snedecor's F distribution and the George W. Snedecor Awards of the American Statistical Association are named after him.

George Waddel Snedecor

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- 6 Inter Relationship Among T, F, χ^2 Variables.

Key Word : F-distribution.

Objectives :

- (1) To derive the probability density function of F-distribution using two independent chi-square variates.
- (2) To study the properties of F-distribution.
- (3) To understand the interrelationship among T, F, χ^2 variables.

4.0 Introduction

Like Student's t-distribution, Snedecor's F-distribution is useful in statistical inference. Such as tests of hypothesis for equality of means of three or more groups, non-linearity of regression, equality of variances, is defined as distribution of ratio of two independent chi-square random variables.

4.1 P.D.F. of F-Distribution

Definition : Suppose U and V are independent chi-square random variables with n_1 and n_2 degrees of freedom respectively then $F = \frac{U/n_1}{V/n_2}$ is said to follow Snedecor's F-distribution with parameters n_1 and n_2 . We symbolically write it as $F_{(n_1, n_2)}$.

Derivation of p.d.f. of $F_{(n_1, n_2)}$:

Since F is a transformation of U and V , we find its p.d.f. as follows:

(i) Let $F = \frac{U/n_1}{V/n_2}$ and $Z = V$ Clearly, $0 < F < \infty$ and $0 < Z < \infty$.

(ii) We get, $U = \frac{n_1}{n_2} FZ$ and $V = Z$.

Hence the Jacobian of transformation

$$J = \frac{\partial(U, V)}{\partial(F, Z)} = \begin{vmatrix} \frac{\partial U}{\partial F} & \frac{\partial U}{\partial Z} \\ \frac{\partial V}{\partial F} & \frac{\partial V}{\partial Z} \end{vmatrix} = \begin{vmatrix} \frac{n_1}{n_2} Z & \frac{n_1}{n_2} F \\ 0 & 1 \end{vmatrix} = \frac{n_1}{n_2} Z$$

(iii) Joint p.d.f. of (F, Z) = Joint p.d.f. of (U, V) . $|J|$

$$g(F, Z) = f(U, V) \cdot |J|$$

$$= f_1(U) \cdot f_2(V) |J| \quad (\because U \text{ and } V \text{ are independent})$$

$$= \left[\left(\frac{1}{2} \right)^{\frac{n_1}{2}} e^{-\frac{U}{2}} \cdot U^{\frac{n_1}{2}-1} \right] \cdot \left[\left(\frac{1}{2} \right)^{\frac{n_2}{2}} e^{-\frac{V}{2}} V^{\frac{n_2}{2}-1} \right] \cdot \left[\frac{n_1}{2} \right]$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \left[\frac{n_1}{2} \right]^{\frac{n_1}{2}} \left[\frac{n_2}{2} \right]^{\frac{n_2}{2}}} \cdot e^{-\left(\frac{U+V}{2}\right)} \cdot U^{\frac{n_1}{2}-1} \cdot V^{\frac{n_2}{2}-1} |J|$$

$$= C \cdot e^{-\frac{1}{2} \left(\frac{n_1}{n_2} FZ + Z \right)} \left(\frac{n_1}{n_2} FZ \right)^{\frac{n_1}{2}-1} \cdot Z^{\frac{n_2}{2}-1} \cdot \frac{n_1}{n_2} Z$$

$$\text{where, } C = \frac{1}{2^{\frac{n_1+n_2}{2}} \left[\frac{n_1}{2} \right]^{\frac{n_1}{2}} \left[\frac{n_2}{2} \right]^{\frac{n_2}{2}}}$$

$$g(f, z) = C \cdot \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} \cdot f^{\frac{n_1}{2}-1} \cdot z^{\frac{n_1+n_2}{2}-1} \cdot e^{-\frac{1}{2} \left(1 + \frac{n_1}{n_2} f \right) z}$$

(iv) To get marginal p.d.f. of F we integrate $g(F, Z)$ over the range of Z .

$$P(f) = C \cdot \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} \cdot f^{\frac{n_1}{2}-1} \int_0^{\infty} e^{-\frac{1}{2} \left(1 + \frac{n_1}{n_2} f \right) z} \cdot z^{\frac{n_1+n_2}{2}-1} \cdot dz$$

$$= C \cdot \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} \cdot f^{\frac{n_1}{2}-1} \cdot \frac{\left[\frac{n_1+n_2}{2} \right]}{\left[\frac{1}{2} \left(1 + \frac{n_1}{n_2} f \right) \right]^{\frac{n_1+n_2}{2}}}$$

$$= \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} \cdot \frac{\left[\frac{n_1+n_2}{2} \right]}{\left[\frac{n_1}{2} \right] \left[\frac{n_2}{2} \right]} \times \frac{f^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} f \right)^{\frac{n_1+n_2}{2}}}$$

$$\therefore P(f) = \frac{\left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}}}{B \left(\frac{n_1}{2}, \frac{n_2}{2} \right)} \times \frac{f^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} f \right)^{\frac{n_1+n_2}{2}}}, F > 0$$

Note : Symbolically we write a r.v. F follows F distribution with n_1 and n_2 d.f. as $F \rightarrow F_{(n_1, n_2)}$.

4.2 Mean, Variance, Moments of $F_{(n_1, n_2)}$

(a) Mean $E(F) = \int_0^{\infty} f \cdot p(f) df$

$$= \int_0^{\infty} \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot F \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}} dF$$

Substituting $x = \frac{n_1}{n_2} F$, $dF = \frac{n_2}{n_1} dx$

$$\begin{aligned} \therefore E(F) &= \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \int_0^{\infty} \frac{x^{\frac{n_1}{2}}}{(1+x)^{\frac{n_1+n_2}{2}}} \cdot \frac{n_2}{n_1} dx \\ &= \frac{n_2}{n_1} \cdot \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \times B\left(\frac{n_1}{2} + 1, \frac{n_2}{2} - 1\right) \quad \text{if } \frac{n_2}{2} > 1 \\ &= \frac{n_2}{n_1} \cdot \frac{\frac{n_1}{2} + 1}{\frac{n_1+n_2}{2}} \times \frac{\frac{n_2}{2} - 1}{\frac{n_1+n_2}{2}} \quad \text{if } \frac{n_2}{2} > 2 \\ &= \frac{n_2}{n_1} \cdot \frac{n_1/2}{(n_2/2 - 1)} = \frac{n_2}{n_1} \cdot \frac{n_1}{(n_2 - 2)} = \frac{n_2}{n_2 - 2} \quad \text{if } \frac{n_2}{2} > 2 \end{aligned}$$

Note :

- Mean $= \frac{n_2}{n_2 - 2}$ which is independent of the other parameter n_1 .
- Mean $= \frac{n_2}{n_2 - 2} > 1$.
- We can verify here that $E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)}$. If U and V are independent chi-square r.v.s. with n_1 and n_2 d.f. respectively then $\frac{U/n_1}{V/n_2}$ follows $F(n_1, n_2)$.

$$\therefore E\left(\frac{U/n_1}{V/n_2}\right) = \frac{n_2}{n_2 - 2} \text{ where as } \frac{E(U/n_1)}{E(V/n_2)} = 1$$

Thus if $X = U/n_1$ and $Y = V/n_2$ then, $E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)} = 1$

(b) r^{th} Raw Moment

$$\begin{aligned} \mu_r' &= E(F^r) = \int_0^{\infty} f^r \cdot p(f) \cdot df \\ &= \int_0^{\infty} \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot f^r \cdot \frac{f^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} f\right)^{\frac{n_1+n_2}{2}}} df \end{aligned}$$

Substituting $\frac{n_1}{n_2} F = x$ $\therefore F = \frac{n_2}{n_1} = x$ we get $dF = \frac{n_2}{n_1} dx$

$$\begin{aligned} \mu_r' &= \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \cdot \int_0^{\infty} \frac{\left(\frac{n_2}{n_1} x\right)^{\frac{n_1}{2} + r - 1}}{(1+x)^{\frac{n_1+n_2}{2}}} \cdot \frac{n_2}{n_1} dx \\ &= \left(\frac{n_2}{n_1}\right)^r \cdot \frac{B\left(\frac{n_1}{2} + r, \frac{n_2}{2} - r\right)}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \quad \text{if } \frac{n_2}{2} > r \end{aligned}$$

In particular for $r = 1$ we get

$$\mu_1' = \frac{n_2}{n_1} \cdot \frac{B\left(\frac{n_1}{2} + 1, \frac{n_2}{2} - 1\right)}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} = \frac{n_2}{n_2 - 2}$$

For $r = 2$, we get

$$\mu_2' = \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{B\left(\frac{n_1}{2} + 2, \frac{n_2}{2} - 2\right)}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \quad \text{if } \frac{n_2}{2} > 2$$

$$= \left(\frac{n_2}{n_1}\right)^2 \cdot \frac{\sqrt{\frac{n_1}{2} + 2} \sqrt{\frac{n_2}{2} - 2}}{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} = \left(\frac{n_2}{n_1}\right)^2 \frac{\left(\frac{n_1}{2} + 1\right) \left(\frac{n_1}{2}\right)}{\left(\frac{n_1}{2} - 1\right) \left(\frac{n_2}{2} - 2\right)} \quad \text{if } n_2 > 4$$

$$= \left(\frac{n_2}{n_1}\right)^2 \frac{n_1 (n_1 + 2)}{(n_2 - 2) (n_2 - 4)} = \frac{n_2^2 (n_1 + 2)}{n_1 (n_2 - 2) (n_2 - 4)}$$

$$\text{Variance} = \mu_2' - (\mu_1')^2 \quad \text{if } n_2 > 4.$$

$$= \frac{n_2^2 (n_1 + 2)}{n_1 (n_2 - 2) (n_2 - 4)} - \left(\frac{n_2}{n_2 - 2}\right)^2 = \frac{n_2^2}{n_2 - 2} \left[\frac{n_1 + 2}{n_1 (n_2 - 4)} - \frac{1}{n_2 - 2} \right]$$

$$= \frac{n_2^2}{n_2 - 2} \left[\frac{n_1 n_2 - 2 n_2 - 2 n_1 - 4 - n_1 n_2 + 4 n_1}{n_1 (n_2 - 2) (n_2 - 4)} \right]$$

$$= \frac{2 n_2^2 (n_1 + n_2 - 2)}{n_1 (n_2 - 2)^2 (n_2 - 4)} \quad \text{if } n_2 > 4.$$

4.3 Mode of $F(n_1, n_2)$

M is mode of the distribution if it is a point of maxima of p.d.f. $P(F)$. It can be obtained by solving $P'(F) = 0$, such that $P''(F) < 0$.

$$P(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2} - 1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1 + n_2}{2}}}$$

$$P(F) = C \cdot \frac{F^{\frac{n_1}{2} - 1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1 + n_2}{2}}}, \text{ where, } C = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)}$$

Taking logarithms on both sides

$$\therefore \log P(F) = \log C + \left(\frac{n_1}{2} - 1\right) \log F - \left(\frac{n_1 + n_2}{2}\right) \log \left(1 + \frac{n_1}{n_2} F\right)$$

Differentiating both the sides w.r.t. F , we get

$$\frac{P'(F)}{P(F)} = 0 + \frac{\frac{n_1}{2} - 1}{F} - \left(\frac{n_1 + n_2}{2}\right) \frac{\frac{n_1}{n_2}}{1 + \frac{n_1}{n_2} F}$$

$$P'(F) = \left\{ \frac{n_1 - 2}{2F} - \frac{n_1 (n_1 + n_2)}{2 n_2} \times \frac{n_2}{n_2 + n_1 F} \right\} P(F) \quad \dots (1)$$

$$\therefore P'(F) = 0, \text{ gives}$$

$$\therefore \frac{n_1 - 2}{2F} = \frac{n_1 (n_1 + n_2)}{2 (n_2 + n_1 F)}$$

$$\therefore (n_2 + n_1 F) (n_1 - 2) = n_1 (n_1 + n_2) F$$

$$\therefore n_1 (n_1 - 2) F - n_1 n_2 F = n_1^2 F - n_2 (n_1 - 2)$$

$$\therefore F = \frac{n_2}{n_2 + 2} \cdot \frac{n_1 - 2}{n_1}; \text{ if } n_1 > 2$$

Differentiating equation (1) w.r.t. F we can verify

$$P''(F) < 0 \text{ at } \frac{n_2 (n_1 - 2)}{n_1 (n_2 + 2)}$$

$$\text{Hence the mode is } \left(\frac{n_2}{n_2 + 2}\right) \left(\frac{n_1 - 2}{n_1}\right).$$

Remark : Note that $\frac{n_2}{n_2 + 2} < 1$ and $\frac{n_1 - 2}{n_1} < 1$. Hence mode is less

than 1.

4.4 Nature of Probability Curve

Note that mean is $\frac{n_2}{n_2 - 2}$ (if $n_2 > 2$).

which is greater than 1. However mode is smaller than 1. Hence $\text{Mean} - \text{Mode} > 0$. Therefore Karl Pearson's coefficient of skewness = $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}} > 0$ for $n_1 > 2$ and $n_2 > 2$. It indicates that the probability curve has positive skewness. As $n_1 \rightarrow \infty$, $n_2 \rightarrow \infty$ the curve becomes symmetric.

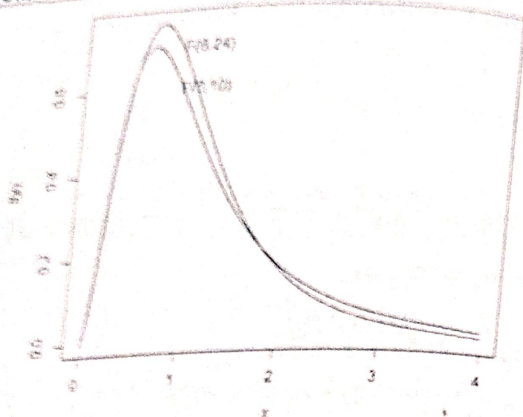


Fig. 4.1 : F-distribution probability curve

Distribution	Mean	Mode
$F_{(8, 24)}$	1.0909	0.6154
$F_{(8, 108)}$	1.25	0.5556

Result (1) : If X follows F-distribution with n_1, n_2 d.f., then $\frac{1}{X}$ follows F distribution with n_2, n_1 d.f.

Solution : Let $Y = \frac{1}{X}$, therefore $Y > 0$.

$$X = \frac{1}{Y}, J = \frac{dx}{dy} = -\frac{1}{Y^2}$$

$\therefore$ p.d.f. Y is given by,

$$g(y) = f(x) |J| \text{ where, } f(x) \text{ is p.d.f. of } X$$

$$\begin{aligned} \therefore g(y) &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{\frac{n_1}{2} - 1}{x^{\frac{n_1}{2} - 1}} \cdot |J| \\ &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \left(\frac{1}{y}\right)^{\frac{n_1}{2} - 1} \cdot \frac{1}{y^2} \\ &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \left(1 + \frac{n_1}{n_2} y\right)^{\frac{n_1 + n_2}{2}} \cdot \frac{1}{y^2} \end{aligned}$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \left(\frac{1}{y}\right)^{\frac{n_1}{2} - 1} \cdot \left[\frac{\frac{n_2 y}{n_1}}{\frac{n_2 y}{n_1} + 1} \right]^{\frac{n_1 + n_2}{2}}$$

[Multiplying Numerator and Denominator by $\frac{n_2 y}{n_1}$]

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \cdot \left(\frac{n_2}{n_1}\right)^{\frac{n_1 + n_2}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{y^{\frac{n_2}{2} - 1}}{\left(1 + \frac{n_2 y}{n_1}\right)^{\frac{n_1 + n_2}{2}}}$$

$$= \frac{\left(\frac{n_2}{n_1}\right)^{\frac{n_2}{2}}}{B\left(\frac{n_2}{2}, \frac{n_1}{2}\right)} \cdot \frac{y^{\frac{n_2}{2} - 1}}{\left(1 + \frac{n_2 y}{n_1}\right)^{\frac{n_1 + n_2}{2}}}; y > 0$$

$$\left[\because B\left(\frac{n_1}{2}, \frac{n_2}{2}\right) = B\left(\frac{n_2}{2}, \frac{n_1}{2}\right) \right]$$

which is the p.d.f. of $F_{(n_2, n_1)}$

Remark : Thus reciprocal of F variate also follows F distribution with d.f. in reverse order).

4.5 Median and Quartiles of $f_{(n, n)}$

Median : If F is a r.v. with F distribution with (n, n) d.f. then the median of F is 1.

By definition m is median if

$$P(F_{(n, n)} \geq m) = P(F_{(n, n)} \leq m) = \frac{1}{2} \quad \dots (1)$$

$$\therefore P\left(\frac{1}{F_{(n, n)}} \leq \frac{1}{m}\right) = \frac{1}{2}$$

Since, $\frac{1}{F_{(n,n)}}$ also follows $F_{(n,n)}$ we get

$$P\left(F_{(n,n)} \leq \frac{1}{m}\right) = \frac{1}{2}$$

$$\therefore P\left(F_{(n,n)} \geq \frac{1}{m}\right) = \frac{1}{2}$$

Thus from equation (1) and (2) we get

$$m = \frac{1}{m} \text{ i.e. } m^2 = 1$$

$\therefore m = 1$ is the median.

Result 2 : If F follows $F_{(n,n)}$ the first and the third quartiles are reciprocals of each other.

Proof : Let Q_1 and Q_3 be the first and the third quartiles respectively.

$$\text{By definition } P(F_{(n,n)} \leq Q_1) = \frac{1}{4}$$

$$\therefore P\left(\frac{1}{F_{(n,n)}} \geq \frac{1}{Q_1}\right) = \frac{1}{4}$$

$$\therefore P\left(F_{(n,n)} \geq \frac{1}{Q_1}\right) = \frac{1}{4}$$

$$\therefore P\left(F_{(n,n)} \leq \frac{1}{Q_1}\right) = \frac{3}{4}$$

However Q_3 is defined as

$$P(F_{(n,n)} \leq Q_3) = \frac{3}{4}$$

Therefore comparing equation (1) and (2), we get

$$\frac{1}{Q_1} = Q_3 \quad \text{or } Q_1 = \frac{1}{Q_3} \quad \text{or } Q_1 Q_3 = 1$$

Result 3 : If $X \rightarrow F_{(n_1, n_2)}$ and $Y \rightarrow F_{(n_2, n_1)}$ then show that $P(X \geq a) + P\left(Y \geq \frac{1}{a}\right) = 1$.

Proof : Let, $P(X \geq a) = P$

$$\therefore P\left(\frac{1}{X} \leq \frac{1}{a}\right) = P$$

$$P\left(Y \leq \frac{1}{a}\right) = P$$

$$P\left(Y \geq \frac{1}{a}\right) = 1 - P$$

From equation (1) and (2) we get

$$P(X \geq a) + P\left(Y \geq \frac{1}{a}\right) = 1.$$

Note : The above result equivalent to

$$P(X \geq a) = P\left(Y \leq \frac{1}{a}\right)$$

$$\text{i.e. } P(F_{(n_1, n_2)} \geq a) = P\left(F_{(n_2, n_1)} \leq \frac{1}{a}\right)$$

Corollary : If $F_{(n_1, n_2, \alpha)}$ is a point such that

$$P[F_{(n_1, n_2)} \geq F_{(n_1, n_2, \alpha)}] = \alpha$$

$$\text{then show that } F_{(n_1, n_2, \alpha)} = \frac{1}{F_{(n_1, n_2, 1-\alpha)}}$$

Proof : $P[F_{(n_1, n_2)} \geq F_{(n_1, n_2, \alpha)}] = \alpha$

$$P\left[\frac{1}{F_{(n_1, n_2)}} \leq \frac{1}{F_{(n_1, n_2, \alpha)}}\right] = \alpha$$

$$P\left[F_{(n_2, n_1)} \leq \frac{1}{F_{(n_1, n_2, \alpha)}}\right] = 1 - \alpha$$

... (1)

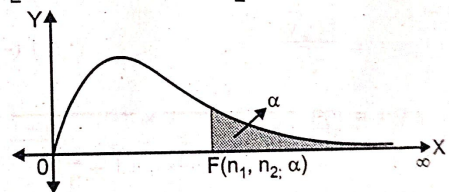


Fig. 4.2

However by definition

$$P[F_{(n_2, n_1)} \geq F_{(n_2, n_1, 1-\alpha)}] = 1 - \alpha$$

... (2)

$$\therefore \text{From equation (1) and (2) we get } F_{(n_1, n_2, \alpha)} = \frac{1}{F_{(n_2, n_1, 1-\alpha)}}$$

4.6 Inter Relationship Among T, F, χ^2 Variables

(A) Relation between T and F.

Note that if U and V are independent r.v.s., $U \rightarrow N(0, 1)$, $V \rightarrow \chi_n^2$ then we get $t = \frac{U}{\sqrt{V/n}}$ as t variate with n d.f. Clearly $t^2 = \frac{U^2}{V/n} = \frac{\chi_1^2 V_1}{\chi_n^2/n} = F_{(1, n)}$. Thus we get if $t \rightarrow t_n$ then $t^2 \rightarrow F_{(1, n)}$.

Result : If $t \rightarrow t_n$, then $t^2 \rightarrow F_{(1, n)}$.

Proof : Let $t \rightarrow t_n$, and $Y = t^2$. We derive distribution function of Y and differentiate it to get p.d.f.

$$\begin{aligned} G(y) &= P(Y \leq y) = P(t^2 \leq y) \\ &= P(-\sqrt{y} \leq t \leq \sqrt{y}) \\ &= P(t \leq \sqrt{y}) - P(t \leq -\sqrt{y}) \\ &= F(\sqrt{y}) - F(-\sqrt{y}) \end{aligned}$$

$F(\cdot)$ is a distribution function of t.

$$\begin{aligned} \therefore \text{p.d.f. of } Y &= g(y) = \frac{d}{dy} G(y) \\ &= \frac{d}{dy} F(\sqrt{y}) - \frac{d}{dy} F(-\sqrt{y}) \\ &= \frac{f(\sqrt{y})}{2\sqrt{y}} - \frac{f(-\sqrt{y})}{2\sqrt{y}} = \frac{f(\sqrt{y}) + f(-\sqrt{y})}{2\sqrt{y}} \end{aligned}$$

where $f(\cdot)$ is a p.d.f. of t

$$= \frac{f(\sqrt{y})}{\sqrt{y}} \quad (\because f(-\sqrt{y}) = f(\sqrt{y}))$$

$$\begin{aligned} \therefore g(y) &= \frac{1}{\sqrt{y}} \cdot \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \times \frac{1}{\left(1 + \frac{y}{n}\right)^{\frac{n+1}{2}}} \\ &= \frac{\left(\frac{1}{n}\right)^{\frac{1}{2}}}{B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{y^{\frac{1}{2}-1}}{\left(1 + \frac{y}{n}\right)^{\frac{n+1}{2}}} \quad (\because y > 0) \end{aligned}$$

which is p.d.f. of $F_{(1, n)}$

(B) Relation between F and χ^2 :

(i) F-distribution is defined as distribution of the ratio of independent chi-square variates divided by their corresponding degrees of freedom.

(ii) If F follows $F_{(n_1, n_2)}$ then probability distribution of $n_1 F$ tends to chi-square probability distribution with n_1 degrees of freedom as $n_2 \rightarrow \infty$.

Proof : Let $Y = n_1 F$, hence $J = \frac{dF}{dY} = \frac{1}{n_1}$

p.d.f. Y = p.d.f. of F |J|

$\therefore$

$$g(y) = \frac{1}{n_1} \cdot \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1 F}{n_2}\right)^{\frac{n_1+n_2}{2}}}$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{\left(\frac{1}{n_1}\right)^{\frac{n_1}{2}} \cdot y^{\frac{n_1}{2}-1}}{\left(1 + \frac{y}{n_2}\right)^{\frac{n_1+n_2}{2}}}$$

$$\lim_{n_2 \rightarrow \infty} g(y) = \lim_{n_2 \rightarrow \infty} \frac{\frac{-n_1}{2} \cdot \frac{n_1+n_2}{2}}{\frac{n_1}{2}} \cdot \frac{y^{\frac{n_1}{2}-1}}{\left(1 + \frac{y}{n_2}\right)^{\frac{n_2}{2}} \left(1 + \frac{y}{n_2}\right)^{\frac{n_1}{2}}}$$

Note that for large n, $\frac{n+k}{n} \approx nk$, k fixed

$$\therefore \lim_{n_2 \rightarrow \infty} \frac{\Gamma\left(\frac{n_1+n_2}{2}\right)}{\Gamma\left(\frac{n_2}{2}\right)} = \left(\frac{n_2}{2}\right)^{\frac{n_1}{2}}$$

$$\text{and } \lim_{a \rightarrow \infty} \left(1 + \frac{k}{a}\right)^a = e^k$$

$$\begin{aligned} \therefore \lim_{n_2 \rightarrow \infty} g(y) &= \lim_{n_2 \rightarrow \infty} \frac{n_2^{-\frac{n_1}{2}} \left(\frac{n_2}{2}\right)^{\frac{n_1}{2}}}{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}}} \frac{y^{\frac{n_1}{2}-1}}{\left(1 + \frac{y}{n_2}\right)^{\frac{n_2}{2}} \left(1 + \frac{y}{n_2}\right)^{\frac{n_1}{2}}} \\ &= \frac{\left(\frac{1}{2}\right)^{\frac{n_1}{2}}}{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}}} \cdot \frac{y^{\frac{n_1}{2}-1}}{e^{y/2}} = \frac{\left(\frac{1}{2}\right)^{\frac{n_1}{2}}}{\left(\frac{n_1}{2}\right)^{\frac{n_1}{2}}} \cdot e^{-y/2} y^{\frac{n_1}{2}-1}; y > 0 \end{aligned}$$

It is p.d.f. of $\chi^2_{n_1}$ r.v. with n_1 d.f.

Solved Examples

Example 4.1 : If X follows $F_{(2, n)}$, then show that

$P(X \geq k) = \left(1 + \frac{2k}{n}\right)^{-\frac{n}{2}}, k > 0$. Also find $\lim_{n \rightarrow \infty} P(X \geq k)$ and compare it with probability of that of a χ^2_2 variate. Thus verify $\lim_{n \rightarrow \infty} P(X \geq k) = P\left(\lim_{n \rightarrow \infty} X \geq k\right)$.

Solution : (i) Since $X \rightarrow F_{(2, n)}$, its p.d.f. will be

$$\begin{aligned} f(x) &= \frac{2}{n} \cdot \frac{1}{B\left(1, \frac{n}{2}\right)} \times \frac{1}{\left(1 + \frac{2x}{n}\right)^{\left(\frac{n+2}{2}\right)}} \\ &= \frac{2}{n} \cdot \frac{\frac{n}{2} + 1}{\frac{n}{2}} \left(1 + \frac{2x}{n}\right)^{-\left(\frac{n+2}{2}\right)} \\ &= \left(1 + \frac{2x}{n}\right)^{-\left(\frac{n+2}{2}\right)} \end{aligned}$$

$$\begin{aligned} \therefore P(X \geq k) &= \int_k^{\infty} \left(1 + \frac{2x}{n}\right)^{-\frac{n+2}{2}} dx = \left[\frac{\left(1 + \frac{2x}{n}\right)^{-\frac{n}{2}}}{-\frac{n}{2}} \times \frac{2}{n} \right]_k^{\infty} \\ &= - \lim_{x \rightarrow \infty} \left(1 + \frac{2x}{n}\right)^{-\frac{n}{2}} + \left(1 + \frac{2k}{n}\right)^{-\frac{n}{2}} \\ &= 0 + \left(1 + \frac{2k}{n}\right)^{-\frac{n}{2}} \end{aligned}$$

$$\lim_{n \rightarrow \infty} P(X \geq k) = \lim_{n \rightarrow \infty} \left(1 + \frac{2k}{n}\right)^{-\frac{n}{2}} = e^{-k}$$

(ii) If $X \rightarrow F_{(n_1, n_2)}$ then as $n_2 \rightarrow \infty, n_1 F \rightarrow \chi^2_{n_1}$

Here $X \rightarrow F_{(2, n)}$, therefore as $n \rightarrow \infty, 2X \rightarrow \chi^2_2$

$$\begin{aligned} \therefore P\left(\lim_{n \rightarrow \infty} X \geq k\right) &= P\left(\lim_{n \rightarrow \infty} 2X \geq 2k\right) \\ &= P(\chi^2_2 \geq 2k) = \int_{2k}^{\infty} \frac{1}{2} e^{-y/2} dy \\ &= \left[-e^{-y/2}\right]_{2k}^{\infty} = -e^{-\infty} + e^{-\frac{2k}{2}} = e^{-k} \end{aligned}$$

Note : Thus from (i) and (ii) we have verified that

$$\lim_{n \rightarrow \infty} P(X \geq k) = P\left(\lim_{n \rightarrow \infty} X \geq k\right) = e^{-k}$$

Example 4.2 : Suppose $X_1, X_2, \dots, X_{25}$ is a random sample from $N(\mu, 10)$ and $Y_1, Y_2, \dots, Y_{21}$ is a random sample from $N(\mu_2, 15)$. If $S_1^2 = \frac{1}{24} \sum (X_i - \bar{X})^2$ and $S_2^2 = \frac{1}{20} \sum (Y_i - \bar{Y})^2$, then find $P\left(\frac{S_1^2}{S_2^2} > 1.3867\right)$.

Solution : If X_i 's are i.i.d. $N(\mu, \sigma^2)$ then $\sum_{i=1}^{25} (X_i - \bar{X})^2 / \sigma^2$ follows χ^2 distribution with 24 d.f.

Points to Remember

1. If χ_1^2 and χ_2^2 are two independent chi-square random variables with n_1 and n_2 degrees of freedom respectively then the ratio $\frac{\chi_1^2/n_1}{\chi_2^2/n_2}$ follows F-distribution with n_1 and n_2 degrees of freedom.
2. The probability density function (p.d.f.) of F-distribution with n_1 and n_2 degrees of freedom is given by

$$P(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}}; F > 0$$

3. If F follows F-distribution with n_1 and n_2 degrees of freedom then

(i) $E(F) = \text{mean} = \frac{n_2}{n_2 - 2} > 1$ which is independent of parameter n_1 .

(ii) The r^{th} raw moment is given by

$$\mu_r' = \left(\frac{n_2}{n_1}\right)^r \frac{B\left(\frac{n_1}{2} + r, \frac{n_2}{2} - r\right)}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \quad \text{if } \frac{n_2}{2} > r.$$

In particular $\mu_2' = \frac{n_2^2 (n_1 + 2)}{n_1 (n_2 - 2) (n_2 - 4)}$ if $n_2 > 4$

(iii) $\text{Var}(F) = \frac{2n_2^2 (n_1 + n_2 - 2)}{n_1 (n_1 - 2)^2 (n_2 - 4)}$ if $n_2 > 4$

(iv) $\text{Mode} = \left(\frac{n_2}{n_2 + 2}\right) \left(\frac{n_1 - 2}{n_1}\right)$ if $n_1 > 2$. It is always less than 1.

(v) $\text{Mean} - \text{Mode} > 0 \therefore$ The density curve is positively skewed.

4. If F is a random variable with (n, n) d.f. then (i) median = 1, (ii) $Q_1 Q_3 = 1$.

5. If $X \rightarrow F(n_1, n_2)$ and $Y \rightarrow F(n_2, n_1)$ then $P(X \geq a) + P\left(Y \geq \frac{1}{a}\right) = 1$ where a is constant.

If X follows F distribution with n_1, n_2 degrees of freedom then $\frac{1}{X}$ follows F-distribution with n_2, n_1 degrees of freedom. Thus reciprocal of F variate also follows F-distribution with d.f. in reverse order.

If $t \rightarrow t_n$ then $t^2 \rightarrow F_{(1, n)}$.

If F follows $F_{(n_1, n_2)}$ then $n_1 F$ tends to chi-square probability distribution with n_1 degrees of freedom as $n_2 \rightarrow \infty$. F-distribution with parameters n, n is symmetric.

If $X_1, X_2, \dots, X_{n_1}$ is a random sample from $N(\mu_1, \sigma^2)$ and $Y_1, Y_2, \dots, Y_{n_2}$ is a random sample from $N(\mu_2, \sigma^2)$ then $\frac{S_1^2}{S_2^2}$ where

$S_1^2 = \frac{\sum (X_i - \bar{X})^2}{n_1 - 1}$ and $S_2^2 = \frac{\sum (Y_i - \bar{Y})^2}{n_2 - 1}$ follows F distribution with $(n_1 - 1)$ and $(n_2 - 1)$ d.f.

Think Over It

1. If X is a continuous r.v. with p.d.f

$$f(x) = \frac{1}{B\left(\frac{1}{2}, \frac{n-2}{2}\right)} (1 - X^2)^{\frac{n-4}{2}}; -1 \leq X \leq 1, n > 2$$

= 0 otherwise

find the probability distribution of $\frac{1+X}{1-X}$.

2. Let X_1, X_2, X_3 be independent normal variates with $E(X_i) = i$ and $\text{var}(X_i) = i^2$; $i = 1, 2, 3$. Using X_1, X_2, X_3 construct a statistic that has F distribution with 1 and 2 d.f.
3. If X_1 and X_2 are i.i.d. r.v.s with pdf $f(x) = e^{-x}$, $x \geq 0$ then show that $\frac{X_1}{X_2}$ follows F distribution. Identify the corresponding degrees of freedom.